



Dashboard 2 Report

Overall SLA Performance & Logistics Health Dashboard

1. Objective

The objective of this dashboard is to provide a consolidated, decision-oriented view of logistics SLA performance across carriers, routes, and shipping modes.

It answers three key business questions:

1. How healthy is our overall SLA performance?
2. Where are the major problem areas (carrier, route, mode)?
3. What should operations and procurement focus on immediately?

This is a descriptive and diagnostic dashboard. No predictive or ML components are included.

2. Data Source

- Dataset: week1_cleaned_data.csv
- Granularity: One row per shipment
- Only cleaned and validated data used
- No feature engineering or predictive metrics

3. KPI Definitions



1 Total Shipments

Definition: Total number of shipments in selected filter context

Calculation:

`COUNT(shipment_id)`

Purpose: Indicates operational volume.



2 SLA Breach Rate (%)

Definition: Percentage of shipments that breached SLA

Calculation:

`SUM(sla_breach_flag) / COUNT(shipment_id)`

Purpose: Primary reliability indicator.



3 On-Time Delivery Rate (%)

Definition: Percentage delivered within SLA

Calculation:

`(COUNT - SUM(sla_breach_flag)) / COUNT`

Purpose: Positive view of service performance.



4 Average Delivery Delay (Days)

Definition: Average delay in days

Includes both positive (late) and negative (early) values.

Purpose: Measures delay severity.



5 Average Shipping Cost (USD)

Definition: Average cost per shipment

Purpose: Supports cost-performance trade-off analysis.

4. Key Visual Insights

◇ SLA Breach Rate by Carrier

Identifies underperforming carriers.

High breach rate indicates potential vendor performance issues.

◇ SLA Breach Rate by Shipping Mode

Highlights risk differences across transport modes (Air, Sea, Road, Rail).

◇ SLA Breach Rate by Route

Matrix view of Origin → Destination combinations.

Helps identify geographic bottlenecks and problematic corridors.

◇ Average Delivery Delay by Carrier

Shows severity of delays beyond just breach frequency.

◇ Cost vs Delivery Delay (Scatter Plot)

Evaluates whether higher shipping cost results in better reliability.

Helps detect inefficiencies in premium shipments.

◇ Shipment Volume Distribution

Shows operational concentration risk.

High-volume + high-breach carriers represent priority action areas.

5. Global Filters

All KPIs and visuals respond dynamically to:

- Carrier
- Shipping Mode
- Region
- Origin Country
- Destination Country
- SLA Breach Flag

This allows interactive drill-down analysis.

6. Business Implications

This dashboard enables leadership to:

- Instantly assess overall SLA health
- Identify risky carriers and routes
- Evaluate cost vs reliability trade-offs
- Prioritize operational improvements
- Support procurement performance discussions